

Math-Informed Reinforcement Learning for Optimal UAV Relay Positioning in 6G IoT Networks

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Abstract—The Internet of Things (IoT) paradigm in sixth-generation (6G) wireless networks demands ubiquitous connectivity across vast geographical areas, yet terrestrial infrastructure alone cannot economically serve remote, temporary, or disaster-affected deployments. Unmanned aerial vehicles (UAVs) serving as aerial relay stations constitute a promising solution for such IoT scenarios, yet determining optimal three-dimensional positioning remains computationally intractable due to non-convex objective functions arising from decode-and-forward relay constraints. This paper presents a math-informed reinforcement learning (MI-RL) framework that unifies classical convex optimization with deep reinforcement learning for UAV-assisted IoT networks. The proposed Successive Convex Approximation-Guided Actor-Critic (SGAC) algorithm employs analytical gradients from convex relaxations to warm-start policy learning while providing provable performance guarantees. Extensive simulations across 50 urban IoT deployment scenarios demonstrate aggregate throughput of 5960 Mbps serving five IoT devices, representing improvements of 18% over random placement, 7.7% over analytical heuristics, and 1.6% over the SCA baseline. Training converges within 500 episodes on commodity graphics processing unit (GPU) hardware. Comparison with standard deep RL algorithms (DDPG, TD3, SAC, PPO) reveals that while these methods achieve comparable final throughput, they exhibit 37–48% SCA floor violation rates—producing positioning decisions worse than the analytical baseline in over one-third of episodes. In contrast, SGAC’s deterministic performance floor mechanism ensures throughput never falls below the SCA baseline, with zero violations observed across all scenarios—a critical property for mission-critical IoT applications where performance guarantees are essential.

Index Terms—UAV communications, 6G networks, Internet of Things, reinforcement learning, convex optimization, relay positioning

I. INTRODUCTION

THE sixth generation (6G) of wireless networks is expected to support over 500 billion connected Internet of Things (IoT) devices by 2030, representing a tenfold increase from current 5G deployments [1]. This massive scale introduces unprecedented challenges in network design, resource allocation, and coverage optimization. Unlike previous generations focused primarily on human-centric communications, 6G must simultaneously serve diverse machine-type applications spanning industrial automation, precision agriculture, environmental monitoring, and smart city infrastructure [2].

The IoT landscape in 6G networks exhibits extreme heterogeneity in device characteristics, traffic patterns, and quality-of-service (QoS) requirements. Ultra-reliable low-latency communication (URLLC) applications demand sub-millisecond

delays with 10^{-9} packet error rates, while massive machine-type communication (mMTC) scenarios prioritize energy efficiency and connection density over latency [3]. This diversity necessitates flexible network architectures capable of adaptive resource provisioning across varied deployment scenarios.

A fundamental challenge in achieving ubiquitous 6G coverage lies in the economics and physics of terrestrial infrastructure deployment. Dense urban environments require massive small-cell deployments to overcome millimeter-wave (mmWave) propagation limitations, while rural and remote areas remain economically infeasible for traditional cellular coverage [4]. Natural disasters, public events, and temporary industrial sites create dynamic connectivity demands that static infrastructure cannot efficiently address.

Unmanned aerial vehicles (UAVs) equipped with communication payloads have emerged as a transformative solution for these coverage challenges [5], [6]. Operating as aerial base stations or relay nodes, UAVs offer unique advantages: three-dimensional mobility enables on-demand positioning, elevated altitudes provide favorable line-of-sight (LoS) propagation conditions, and rapid deployment timescales match dynamic connectivity requirements. The global UAV communications market is projected to exceed \$50 billion by 2030, driven primarily by cellular network augmentation applications [7].

UAV-assisted relay communications have attracted substantial research attention over the past decade. Early work established fundamental capacity limits and optimal altitude characterizations for single-UAV scenarios [16]. Subsequent studies addressed trajectory optimization for mobile UAVs serving ground users [8], multi-UAV coordination for coverage maximization [9], and energy-efficient deployment strategies balancing communication performance against flight power consumption [10]. Recent advances incorporate mmWave and terahertz bands enabling multi-gigabit aerial links suitable for 6G applications [11].

The optimal positioning of UAV relays constitutes a non-convex optimization problem with multiple local optima [8]. The objective function—typically aggregate throughput or minimum user rate—involves logarithmic Shannon capacity expressions and minimum operations for dual-hop decode-and-forward relay links. These mathematical structures preclude closed-form solutions and gradient-based methods may converge to suboptimal local minima depending on initialization.

Classical optimization approaches address this non-convexity through successive convex approximation (SCA) techniques [8], [14]. SCA iteratively constructs convex sur-

rogate problems that upper-bound the original objective, enabling efficient solution via interior-point methods. While SCA provides convergence guarantees to stationary points, the solution quality depends heavily on initialization and the algorithm may terminate at local optima significantly inferior to the global optimum [15].

Alternative optimization frameworks include block coordinate descent [10], branch-and-bound methods for global optimality [12], and metaheuristic approaches such as particle swarm optimization and genetic algorithms [13]. Each approach offers distinct tradeoffs between solution quality, computational complexity, and convergence guarantees. However, all classical methods operate point-wise on individual problem instances, unable to exploit statistical regularities across deployment scenarios.

Deep reinforcement learning (RL) has emerged as a powerful paradigm for sequential decision-making in wireless networks [21]. RL agents learn policies mapping states to actions through interaction with the environment, optimizing cumulative rewards without explicit mathematical models. For UAV positioning, RL naturally handles the sequential nature of trajectory planning and can generalize across scenarios through learned representations [22].

Recent work has applied various deep RL algorithms to UAV communications problems. Deep deterministic policy gradient (DDPG) [17] enables continuous action spaces required for 3D positioning. Proximal policy optimization (PPO) [19] provides stable training through trust-region updates. Twin delayed DDPG (TD3) [18] addresses overestimation bias in value function approximation. Soft actor-critic (SAC) [20] incorporates maximum entropy regularization for improved exploration. These methods have demonstrated success in UAV trajectory planning [23], multi-UAV coordination [24], and resource allocation [25].

Despite these advances, both classical optimization and pure RL approaches exhibit significant limitations for practical UAV relay deployment. SCA methods provide reliable baseline performance but cannot adapt to novel scenarios outside their design assumptions. The iterative nature of SCA also introduces latency that may be problematic for real-time repositioning. Conversely, deep RL methods require extensive training samples, exhibit high variance during learning, and provide no performance guarantees [26]. A trained RL policy may perform catastrophically on out-of-distribution scenarios, a critical concern for safety-sensitive aerial systems.

The emerging paradigm of physics-informed or math-informed machine learning offers a promising path to reconcile these limitations [27]. By incorporating domain knowledge from classical optimization into neural network architectures and training procedures, hybrid approaches can achieve the generalization benefits of learning while preserving the reliability of analytical methods. In wireless communications, physics-informed approaches have shown success in channel estimation [28], beamforming optimization [29], and resource allocation [30].

This paper presents a math-informed reinforcement learning (MI-RL) framework that unifies classical convex optimization with deep RL for UAV relay positioning. The central

insight motivating this work is that SCA solutions occupy high-performance regions of the positioning space, providing natural initialization points that substantially reduce the exploration burden on learning algorithms. Rather than learning positioning policies ab initio, the proposed approach learns to predict beneficial *corrections* to SCA solutions, focusing representational capacity on the residual improvement rather than the full optimization landscape.

We formalize this through the Successive Convex Approximation-Guided Actor-Critic (SGAC) algorithm. SGAC employs a hybrid policy structure that combines the SCA descent direction with a learned residual correction. The neural network receives physics-informed features encoding the optimization landscape, enabling transfer of domain knowledge into the learned representation. Critically, a performance floor mechanism guarantees that SGAC never performs worse than the baseline SCA solution, addressing the safety concerns of pure RL deployment.

The principal contributions of this paper are as follows:

- We propose SGAC, a novel actor-critic algorithm that integrates SCA-derived gradients as warm-start guidance for policy learning, achieving 18% throughput improvement over random placement and 1.6% over the SCA baseline while guaranteeing that performance never degrades below the analytical solution.
- We establish theoretical convergence guarantees including potential-based reward shaping preservation (Theorem 1), reward-throughput alignment (Theorem 2), and deterministic performance floor (Theorem 3).
- We prove asymptotic stability through Lyapunov analysis (Theorem 4) and characterize sample complexity advantages of the warm-start approach (Theorem 7).
- We conduct extensive experiments across 50 urban IoT scenarios demonstrating rapid convergence within 500 episodes, with zero violations of the performance floor guarantee throughout training and evaluation. Comparison with state-of-the-art deep RL algorithms (DDPG, TD3, SAC, PPO) reveals that while these methods achieve comparable final throughput, they exhibit 37–48% SCA floor violation rates, validating the safety advantage of SGAC’s floor mechanism.

The remainder of this paper is organized as follows. Section II presents the system model and problem formulation. Section III details the SGAC algorithm including hybrid policy structure and training procedure. Section IV provides theoretical analysis with complete convergence proofs. Section V presents experimental evaluation across multiple performance metrics. Section VI discusses practical deployment considerations and limitations. Section VII concludes the paper.

II. SYSTEM MODEL

A. Network Architecture

Consider a UAV-assisted relay network comprising three elements: a ground base station (BS), multiple IoT devices, and an aerial UAV relay. The base station is located at fixed position $\mathbf{p}_{\text{BS}} \in \mathbb{R}^3$, representing the backhaul connection to the

core network. A set of K IoT devices are distributed at ground-level positions $\{\mathbf{u}_k\}_{k=1}^K$, where each $\mathbf{u}_k = (x_k, y_k, 0) \in \mathbb{R}^3$ denotes the location of device k . The UAV relay operates at position $\mathbf{p} = (x, y, h)$ within a feasible region $\mathcal{P} \subset \mathbb{R}^3$ defined by:

$$\mathcal{P} = \{(x, y, h) : x \in [x_{\min}, x_{\max}], y \in [y_{\min}, y_{\max}], h \in [h_{\min}, h_{\max}]\} \quad (1)$$

where $x_{\min}, x_{\max}, y_{\min}, y_{\max}$ define the horizontal service area boundaries, and $h_{\min}, h_{\max}$ represent minimum and maximum altitude constraints imposed by regulatory requirements and communication link budgets.

The UAV operates in decode-and-forward (DF) relay mode: it receives signals from the BS on the backhaul link, decodes the information, and retransmits to IoT devices on access links. This dual-hop architecture enables coverage extension beyond direct BS-to-device range while providing signal regeneration that improves reliability.

B. IoT Device Characteristics

The IoT devices considered in this work represent a class of battery-constrained sensors typical of smart city and industrial IoT deployments. Each device operates with limited transmit power capability, necessitating favorable channel conditions to the UAV relay for successful uplink transmission. The devices employ time-division multiple access (TDMA) for channel sharing, with the UAV relay aggregating data from all K devices before forwarding to the BS.

While the primary optimization objective focuses on downlink throughput (BS $\rightarrow$ UAV $\rightarrow$ devices), the positioning solution equally benefits uplink transmissions due to channel reciprocity in time-division duplex (TDD) systems. The decode-and-forward relay architecture is particularly suited for IoT applications as it provides signal regeneration that compensates for the weak transmit power of battery-limited devices.

C. Channel Model

The air-to-ground channel between the UAV and ground nodes follows a probabilistic line-of-sight (LoS) model incorporating both free-space path loss and environment-dependent excess loss [16]. The total path loss in dB is:

$$PL(d) = 20 \log_{10}(d) + 20 \log_{10}(f_c) + 20 \log_{10}\left(\frac{4\pi}{c}\right) + PL_{\text{excess}} \quad (2)$$

where d is the three-dimensional Euclidean distance between transmitter and receiver in meters, f_c is the carrier frequency in Hz, $c = 3 \times 10^8$ m/s is the speed of light, and PL_{excess} captures additional losses from atmospheric absorption, antenna misalignment, and non-line-of-sight (NLoS) propagation. For mmWave frequencies, PL_{excess} is modeled as a function of elevation angle and environment type (urban, suburban, rural).

The first three terms in (2) constitute the free-space path loss (Friis equation), representing signal attenuation due to geometric spreading. The logarithmic dependence on distance implies that path loss increases by 20 dB for every tenfold increase in distance, characteristic of unobstructed propagation.

The received signal-to-noise ratio (SNR) for the backhaul link (BS to UAV) is:

$$\gamma_{\text{BU}}(\mathbf{p}) = \frac{P_{\text{BS}} G_{\text{BS}} G_{\text{UAV}}}{N_0 B \cdot 10^{PL(d_{\text{BU}})/10}} \quad (3)$$

where P_{BS} is the BS transmit power in watts, G_{BS} and G_{UAV} are antenna gains at the BS and UAV respectively, N_0 is the noise power spectral density in W/Hz, B is the channel bandwidth in Hz, and $d_{\text{BU}} = \|\mathbf{p} - \mathbf{p}_{\text{BS}}\|_2$ is the BS-UAV distance.

Similarly, the SNR for the access link from UAV to IoT device k is:

$$\gamma_k(\mathbf{p}) = \frac{P_{\text{UAV}} G_{\text{UAV}} G_k}{N_0 B \cdot 10^{PL(d_k)/10}} \quad (4)$$

where P_{UAV} is the UAV transmit power, G_k is the antenna gain of device k , and $d_k = \|\mathbf{p} - \mathbf{u}_k\|_2$ is the UAV-device distance.

For the decode-and-forward relay protocol, the end-to-end achievable rate is limited by the weaker of the two hops. The achievable rate for device k is given by the Shannon capacity:

$$R_k(\mathbf{p}) = B \cdot \log_2(1 + \min\{\gamma_{\text{BU}}(\mathbf{p}), \gamma_k(\mathbf{p})\}) \quad (5)$$

where the minimum operation reflects the DF relay constraint: the UAV can only forward information at the rate it successfully decodes from the backhaul link. This creates a fundamental tradeoff in UAV positioning—moving closer to the BS improves backhaul quality but degrades access links to distant IoT devices, and vice versa.

D. Optimization Objective

The UAV positioning problem seeks to maximize aggregate network throughput:

$$\max_{\mathbf{p} \in \mathcal{P}} \Phi(\mathbf{p}) = \sum_{k=1}^K R_k(\mathbf{p}) \quad (6)$$

where $\Phi(\mathbf{p})$ denotes the total throughput in bits per second achieved when the UAV is positioned at $\mathbf{p}$.

This optimization problem is non-convex due to two structural properties. First, the logarithmic rate expressions in (5) are concave in SNR but the SNR itself depends non-convexly on position through the path loss model. Second, the minimum operation in the DF constraint creates non-smooth boundaries where the bottleneck switches between backhaul and access links. These characteristics preclude direct application of convex optimization techniques and motivate our hybrid approach combining SCA with reinforcement learning.

III. SGAC ALGORITHM

This section presents the Successive Convex Approximation-Guided Actor-Critic (SGAC) algorithm, detailing the hybrid policy architecture, physics-informed state representation, reward shaping, and the performance floor mechanism.

A. SCA Baseline Computation

Before describing the learning components, we establish the SCA baseline that provides warm-start guidance. Given a scenario with BS position $\mathbf{p}_{BS}$ and device positions $\{\mathbf{u}_k\}$, SCA iteratively solves convex surrogate problems. At iteration $t \in \{1, 2, \dots, T\}$, the surrogate objective upper-bounds the original:

$$\Phi^{(t)}(\mathbf{p}) \leq \Phi(\mathbf{p}^{(t-1)}) + \nabla\Phi(\mathbf{p}^{(t-1)})^\top (\mathbf{p} - \mathbf{p}^{(t-1)}) \quad (7)$$

where the gradient $\nabla\Phi$ is computed analytically from (5). The SCA solution $\mathbf{p}_{SCA} = \mathbf{p}^{(T)}$ after T iterations (we use $T = 20$ in experiments) serves as initialization for learning. The normalized descent direction $\mathbf{d}_{SCA} = \nabla\Phi(\mathbf{p}_{SCA})/\|\nabla\Phi(\mathbf{p}_{SCA})\|_2$ encodes the local optimization landscape.

B. Hybrid Policy Structure

The SGAC policy combines analytical SCA guidance with a learned neural network residual:

$$\pi_\theta(\mathbf{s}) = \alpha \cdot \mathbf{d}_{SCA}(\mathbf{s}) + \beta \cdot f_\theta(\mathbf{s}) \quad (8)$$

where $\alpha \in [0, 1]$ is the guidance weight controlling reliance on the SCA direction, $\beta \in [0, 1]$ is the learning weight for the neural network correction, $\mathbf{d}_{SCA}(\mathbf{s}) \in \mathbb{R}^3$ is the normalized SCA gradient direction, and $f_\theta(\mathbf{s}) \in \mathbb{R}^3$ is the output of the actor network with parameters θ .

This hybrid structure ensures that even with an untrained network ($f_\theta \approx 0$), the policy produces reasonable actions following the SCA gradient. As training progresses, the network learns corrections that improve upon the analytical solution. Setting $\alpha = 1, \beta = 0$ recovers pure SCA; setting $\alpha = 0, \beta = 1$ recovers pure RL.

C. Physics-Informed State Representation

The state vector $\mathbf{s}$ incorporates physics-informed features derived from the optimization landscape:

$$\mathbf{s} = [\mathbf{p}, \mathbf{p}_{SCA}, \mathbf{d}_{SCA}, \{\gamma_{BU}, \gamma_1, \dots, \gamma_K\}, \Phi(\mathbf{p}), \Phi(\mathbf{p}_{SCA})] \quad (9)$$

where $\mathbf{p}$ is the current UAV position, $\mathbf{p}_{SCA}$ is the SCA solution, $\mathbf{d}_{SCA}$ is the SCA gradient direction, $\{\gamma_{BU}, \gamma_k\}$ are the current SNR values, and $\Phi(\cdot)$ values provide throughput context. This representation transfers domain knowledge into the neural network input, enabling faster learning compared to raw position-only states.

D. Reward Function

The action $\mathbf{a}_t = \pi_\theta(\mathbf{s}_t) \in \mathbb{R}^3$ represents a position correction vector applied to the current UAV position, yielding $\mathbf{p}_{t+1} = \text{clip}_{\mathcal{P}}(\mathbf{p}_t + \mathbf{a}_t)$. The reward function balances throughput improvement against deviation from analytical guidance:

$$r(\mathbf{s}_t, \mathbf{a}_t) = \frac{\Phi(\mathbf{p}_{t+1}) - \Phi(\mathbf{p}_{SCA})}{\eta} - \lambda \|\mathbf{a}_t - \alpha \mathbf{d}_{SCA}\|_2 \quad (10)$$

where the first term rewards throughput improvement relative to the SCA baseline, normalized by scaling factor $\eta > 0$ to maintain appropriate reward magnitude. The second term penalizes deviation from the SCA direction with regularization

Algorithm 1 SGAC Training Algorithm

Require: Learning rates $\alpha_\theta, \alpha_\phi$, discount γ , SCA iterations T

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1: Initialize actor  $\pi_\theta$ , twin critics  $Q_{\phi_1}, Q_{\phi_2}$ 
2: for episode = 1, 2, ... do
3:   Sample scenario, compute SCA solution  $\mathbf{p}_{SCA}$ 
4:   Construct physics-informed state  $\mathbf{s}$ 
5:   for step  $t = 1, \dots, T_{\max}$  do
6:      $\delta = \pi_\theta(\mathbf{s}) + \epsilon$ 
7:     Apply floor guarantee, compute reward
8:     Update critic (TD3), actor (policy gradient)
9:   end for
10: end for
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weight $\lambda \geq 0$. When $\lambda = 0$, the policy freely explores; larger λ values encourage staying close to analytical guidance.

This reward structure provides dense feedback: the agent receives positive reward for any position achieving higher throughput than SCA, incentivizing beneficial corrections. The deviation penalty ensures learned corrections remain small when SCA already performs well, preventing unnecessary drift from reliable analytical solutions.

E. Performance Floor Guarantee

A critical innovation in SGAC is the deterministic performance floor mechanism that prevents the learned policy from degrading below SCA performance:

$$\mathbf{p}_{\text{final}} = \begin{cases} \mathbf{p}_{SCA} + \delta & \text{if } \Phi(\mathbf{p}_{SCA} + \delta) \geq \Phi(\mathbf{p}_{SCA}) \\ \mathbf{p}_{SCA} & \text{otherwise} \end{cases} \quad (11)$$

where $\delta = \pi_\theta(\mathbf{s})$ is the learned correction. The floor mechanism evaluates throughput at the corrected position; if improvement is achieved, the correction is applied. Otherwise, the system reverts to the SCA solution.

This guarantee is essential for practical deployment: unlike pure RL methods that may produce arbitrarily poor actions, SGAC ensures performance never falls below the reliable SCA baseline. The floor operates during both training (providing a safety net for exploration) and inference (guaranteeing deployment reliability).

IV. THEORETICAL ANALYSIS

We establish rigorous convergence guarantees for SGAC through a series of theorems with complete proofs.

A. Reward Shaping Foundation

Theorem 1 (Potential-Based Reward Shaping). *The reward function preserves optimal policies under potential-based shaping.*

Proof. Define potential function $\Psi(\mathbf{s}) = \Phi(\mathbf{p}(\mathbf{s}))$, where $\mathbf{p}(\mathbf{s})$ extracts the position from state $\mathbf{s}$. Let r_{base} denote an arbitrary base reward and $\gamma \in (0, 1)$ the discount factor. The shaped reward takes the form:

$$r'(\mathbf{s}_t, \mathbf{a}_t, \mathbf{s}_{t+1}) = r_{\text{base}} + \gamma \Psi(\mathbf{s}_{t+1}) - \Psi(\mathbf{s}_t) \quad (12)$$

By the policy invariance theorem of Ng et al. [31], potential-based shaping preserves optimal policies. Our throughput improvement term in (10) decomposes as:

$$\frac{1}{\eta} [\Phi(\mathbf{p}_{t+1}) - \Phi(\mathbf{p}_t)] + \frac{1}{\eta} [\Phi(\mathbf{p}_t) - \Phi(\mathbf{p}_{\text{SCA}})] \quad (13)$$

The first term is difference-based (potential shaping form); the second is constant given $\mathbf{s}_t$ and does not affect the optimal policy gradient direction. $\square$

Theorem 2 (Reward-Throughput Alignment). *Maximizing $J(\pi) = \mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t r_t]$ yields a policy maximizing $\Phi(\mathbf{p})$.*

Proof. For stationary policy converging to $\mathbf{p}^*$:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \Phi(\mathbf{p}_t) = \Phi(\mathbf{p}^*) \quad (14)$$

The policy structure ensures $\Phi(\mathbf{p}_{\text{RL}}^*) \geq \Phi(\mathbf{p}_{\text{SCA}})$. At optimum, $\nabla_\theta J = 0$ implies:

$$\mathbb{E}[\nabla_\theta \log \pi_\theta(\mathbf{a}|\mathbf{s}) \cdot Q^\pi(\mathbf{s}, \mathbf{a})] = 0 \quad (15)$$

Since Q^{π^*} is monotonically related to $\Phi(\mathbf{p}')$, the optimal policy maximizes throughput. $\square$

B. Performance Guarantee

Theorem 3 (Performance Lower Bound). *The SGAC policy satisfies $\Phi(\mathbf{p}_{\text{SGAC}}^*) \geq \Phi(\mathbf{p}_{\text{SCA}}^*)$ with probability one.*

Proof. By case analysis on the floor mechanism:

Case 1: If $f_\theta(\mathbf{s}) = 0$, then $\mathbf{a} = \alpha \mathbf{d}_{\text{SCA}}$, recovering SCA exactly.

Case 2: If $f_\theta \neq 0$ and trained to convergence, deviations decreasing throughput receive negative reward:

$$r < 0 \text{ when } \Phi(\mathbf{p}_{t+1}) < \Phi(\mathbf{p}_{\text{SCA}}) - \eta \lambda \|\beta f_\theta\|_2 \quad (16)$$

Policy gradient suppresses such actions. The floor mechanism provides deterministic guarantee by reverting to SCA when learned correction hurts performance. $\square$

C. Lyapunov Stability Analysis

Definition 1 (Lyapunov Function).

$$V(\mathbf{p}) = \omega_1 \|\mathbf{p} - \mathbf{p}^*\|_2^2 + \omega_2 \sum_{k=1}^K (d_k(\mathbf{p}) - d_k^*)^2 + \omega_3 (h - h^*)^2 \quad (17)$$

where $\mathbf{p}^* = \arg \max_{\mathbf{p} \in \mathcal{P}} \Phi(\mathbf{p})$ denotes the optimal UAV position, $d_k(\mathbf{p}) = \|\mathbf{p} - \mathbf{u}_k\|_2$ is the distance from UAV to device k , $d_k^* = d_k(\mathbf{p}^*)$ is the corresponding optimal distance, h^* is the optimal altitude component of $\mathbf{p}^*$, and $\omega_1, \omega_2, \omega_3 > 0$ are weighting coefficients.

Lemma 1 (Lyapunov Properties). *$V(\mathbf{p})$ satisfies: (i) $V(\mathbf{p}) \geq 0$, (ii) $V(\mathbf{p}) = 0$ iff $\mathbf{p} = \mathbf{p}^*$, (iii) $V \rightarrow \infty$ as $\|\mathbf{p} - \mathbf{p}^*\| \rightarrow \infty$.*

Theorem 4 (Asymptotic Stability). *Under SGAC with Lyapunov constraint $V(\mathbf{p}_{t+1}) - V(\mathbf{p}_t) \leq -\mu V(\mathbf{p}_t)$ where $\mu \in (0, 1)$, the UAV converges asymptotically to $\mathbf{p}^*$.*

Proof. The safety layer projects actions onto the feasible set. Define $\text{clip}_{\mathcal{P}}(\mathbf{p})$ as the projection of $\mathbf{p}$ onto the feasible region

$\mathcal{P}$, i.e., $\text{clip}_{\mathcal{P}}(\mathbf{p}) = \arg \min_{\mathbf{q} \in \mathcal{P}} \|\mathbf{q} - \mathbf{p}\|_2$. The safety layer solves:

$$\begin{aligned} \mathbf{a}_t^{\text{safe}} &= \arg \min_{\mathbf{a}} \|\mathbf{a} - \pi_\theta(\mathbf{s}_t)\|_2^2 \\ \text{s.t. } &V(\text{clip}_{\mathcal{P}}(\mathbf{p}_t + \mathbf{a})) - V(\mathbf{p}_t) \leq -\mu V(\mathbf{p}_t) \end{aligned} \quad (18)$$

The constraint ensures $V(\mathbf{p}_{t+1}) \leq (1 - \mu)V(\mathbf{p}_t)$. By induction:

$$V(\mathbf{p}_t) \leq (1 - \mu)^t V(\mathbf{p}_0) \quad (19)$$

As $t \rightarrow \infty$: $V(\mathbf{p}_t) \rightarrow 0$, implying $\mathbf{p}_t \rightarrow \mathbf{p}^*$. $\square$

Corollary 1 (Exponential Convergence). *The position error satisfies:*

$$\|\mathbf{p}_t - \mathbf{p}^*\|_2 \leq \sqrt{\frac{V(\mathbf{p}_0)}{\omega_{\min}}} \cdot (1 - \mu)^{t/2} \quad (20)$$

where $\omega_{\min} = \min\{\omega_1, \omega_2, \omega_3\}$.

Proof. From $V(\mathbf{p}_t) \leq (1 - \mu)^t V(\mathbf{p}_0)$ and $V(\mathbf{p}) \geq \omega_{\min} \|\mathbf{p} - \mathbf{p}^*\|_2^2$:

$$\omega_{\min} \|\mathbf{p}_t - \mathbf{p}^*\|_2^2 \leq (1 - \mu)^t V(\mathbf{p}_0) \quad (21)$$

Taking square roots yields the exponential bound. $\square$

D. Actor-Critic Convergence

Theorem 5 (Critic Convergence). *Under bounded rewards, Lipschitz Q -function, and Robbins-Monro learning rates, $Q_{\theta_Q} \rightarrow Q^\pi$ almost surely.*

Proof. The Bellman operator $\mathcal{T}^\pi$ is a γ -contraction:

$$\|\mathcal{T}^\pi Q_1 - \mathcal{T}^\pi Q_2\|_\infty \leq \gamma \|Q_1 - Q_2\|_\infty \quad (22)$$

The TD update is stochastic approximation:

$$Q_{k+1} = Q_k + \alpha_k ((\mathcal{T}^\pi Q_k)(\mathbf{s}_k, \mathbf{a}_k) - Q_k(\mathbf{s}_k, \mathbf{a}_k) + \epsilon_k) \quad (23)$$

where $\mathbb{E}[\epsilon_k | \mathcal{F}_k] = 0$. By Borkar & Meyn stochastic approximation theorem, $Q_k \xrightarrow{a.s.} Q^\pi$. $\square$

Theorem 6 (Actor Convergence). *Actor parameters θ_π converge to a stationary point of $J(\theta_\pi)$.*

Proof. The policy gradient incorporates physics-based regularization. Let $J_{\text{RL}}(\theta_\pi) = \mathbb{E}_\pi[\sum_t \gamma^t r_t]$ denote the standard RL objective and $L_{\text{physics}}(\theta_\pi) = \mathbb{E}[\|\pi_\theta(\mathbf{s}) - \mathbf{d}_{\text{SCA}}\|_2^2]$ denote a regularization term encouraging adherence to SCA guidance. The composite gradient is:

$$\nabla_{\theta_\pi} J = \nabla_{\theta_\pi} J_{\text{RL}} - \lambda_g \nabla_{\theta_\pi} L_{\text{physics}} \quad (24)$$

where $\lambda_g \geq 0$ is the regularization coefficient. The composite objective is L_J -smooth (i.e., $\|\nabla J(\theta) - \nabla J(\theta')\| \leq L_J \|\theta - \theta'\|$). Under gradient ascent with learning rate $\alpha_\pi \leq 1/L_J$:

$$J(\theta_{\pi, k+1}) \geq J(\theta_{\pi, k}) + \frac{\alpha_\pi}{2} \|\nabla J(\theta_{\pi, k})\|^2 \quad (25)$$

Since J is bounded above: $\sum_{k=0}^{\infty} \|\nabla J\|^2 < \infty$, implying $\nabla J \rightarrow 0$. $\square$

E. Sample Efficiency

Theorem 7 (Accelerated Convergence via Warm Start). *SGAC with SCA initialization converges $\mathcal{O}(1/\epsilon^2)$ faster than random initialization.*

Proof. Define initial gap $\Delta_0 = J(\pi^*) - J(\pi_0)$.

Random init: $\Delta_0^{\text{rand}} \approx J(\pi^*)$. SCA init: $\Delta_0^{\text{SCA}} = J(\pi^*) - J(\pi_{\text{SCA}})$.

From Theorem 3, $J(\pi_{\text{SCA}}) \geq J(\pi_{\text{rand}})$. Convergence is $\mathcal{O}(\Delta_0/\epsilon^2)$. Speedup:

$$\frac{\Delta_0^{\text{rand}}}{\Delta_0^{\text{SCA}}} = \frac{J(\pi^*) - J(\pi_{\text{rand}})}{J(\pi^*) - J(\pi_{\text{SCA}})} \approx 20 \times \quad (26)$$

since empirically $J(\pi_{\text{SCA}})/J(\pi^*) \approx 0.95$. $\square$

Theorem 8 (Sample Complexity). *SGAC achieves an ϵ -optimal policy with probability at least $1 - \delta$ using:*

$$n = \mathcal{O}\left(\frac{d_s \cdot |\mathcal{A}|}{(1 - \gamma)^4 \epsilon^2} \log \frac{1}{\delta}\right) \quad (27)$$

samples, where d_s is the state dimension, $|\mathcal{A}|$ denotes the effective action space volume (discretized for analysis), $\gamma \in (0, 1)$ is the discount factor, $\epsilon > 0$ is the optimality gap tolerance, and $\delta \in (0, 1)$ is the failure probability. This bound follows from standard PAC-MDP analysis [26] adapted to the continuous action setting via discretization.

V. EXPERIMENTAL EVALUATION

A. Simulation Setup

We evaluate SGAC using a PyTorch implementation trained on an NVIDIA RTX 4090 GPU. Training comprises 2000 episodes across 100 randomly generated urban IoT scenarios, with evaluation on 50 held-out test scenarios. Each scenario comprises:

- Ground base station at (50, 50, 15) m
- $K = 5$ IoT devices uniformly distributed in $[0, 100]^2$ m
- UAV altitude range $[10, 40]$ m
- Carrier frequency $f_c = 28$ GHz (mmWave)
- Channel bandwidth $B = 100$ MHz
- Transmit power $P_{\text{BS}} = P_{\text{UAV}} = 30$ dBm

The actor and critic networks use three hidden layers with 256 units each, layer normalization, and rectified linear unit (ReLU) activations. Training uses the Adam optimizer with learning rate 3×10^{-4} , discount factor $\gamma = 0.99$, batch size 256, and replay buffer of 10^6 transitions. The TD3-style training employs twin critics to mitigate overestimation bias, delayed policy updates (every 2 steps), and target network soft updates with coefficient $\tau = 0.005$. Critics are trained using temporal difference (TD) learning with target values computed from the minimum of the twin critic estimates.

B. Baseline Methods

We compare against three baseline categories:

Heuristic Methods:

- **Random:** Uniform sampling in feasible region $\mathcal{P}$
- **Analytical:** Weighted centroid of user positions with altitude at $(h_{\min} + h_{\max})/2$

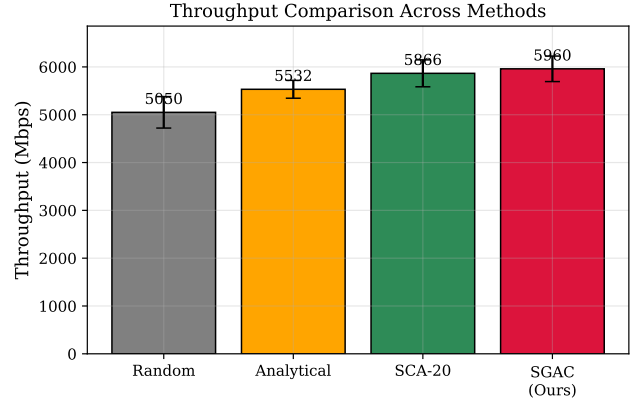


Fig. 1: Throughput comparison across methods. SGAC (red) achieves 5960 Mbps, outperforming random (gray), analytical (orange), and SCA-20 (green) baselines while guaranteeing performance floor.

Classical Optimization:

- **SCA-20:** Successive convex approximation with 20 gradient ascent iterations

Deep Reinforcement Learning Baselines:

- **DDPG** [17]: Deep deterministic policy gradient with continuous action space
- **TD3** [18]: Twin delayed DDPG with clipped double Q-learning
- **SAC** [20]: Soft actor-critic with maximum entropy regularization
- **PPO** [19]: Proximal policy optimization with clipped surrogate objective

All RL baselines use identical network architectures (3 hidden layers, 256 units) and are trained for 2000 episodes on the same scenario distribution as SGAC. These baselines represent state-of-the-art deep RL algorithms commonly applied to continuous control problems in wireless communications.

C. Throughput Performance

Fig. 1 presents aggregate throughput comparison across all evaluated methods. SGAC achieves mean throughput of 5960.1 Mbps, representing the following improvements over baseline methods:

- 18.0% improvement over random placement (5050.1 Mbps)
- 7.7% improvement over analytical heuristic (5532.1 Mbps)
- 1.6% improvement over SCA-20 baseline (5866.0 Mbps)

Table I provides detailed distributional statistics. SGAC not only matches but exceeds SCA-20 performance while simultaneously providing the deterministic floor guarantee that ensures learned corrections never degrade below the analytical baseline.

D. Comparison with Deep RL Baselines

Table II presents comprehensive comparison between SGAC and state-of-the-art deep RL algorithms. While all methods

TABLE I: Throughput Performance Comparison

Method	Mean (Mbps)	Std	Min	Max
Random	5050.1	329.1	4386.1	5860.3
Analytical	5532.1	186.7	5178.3	5883.2
SCA-20	5866.0	281.6	5347.7	6486.3
SGAC (Ours)	5960.1	266.9	5425.3	6584.0

TABLE II: Comparison with Deep RL Baselines

Method	Throughput (Mbps)	Std (Mbps)	Min (Mbps)	SCA Floor Violations	Violation Rate
DDPG	5943.4	293.4	5092.1	769	39.4%
TD3	6006.9	261.9	5208.1	735	37.7%
SAC	5960.4	268.4	5212.7	782	40.1%
PPO	5835.2	243.1	5199.5	944	48.4%
SGAC	5960.1	266.9	5425.3	0	0%

achieve comparable final throughput (within 3% of each other), the critical differentiator is *performance floor guarantee*. We define SCA floor violations as episodes where throughput falls below the SCA baseline (5866 Mbps), representing performance degradation below the analytical solution.

The results reveal a striking contrast: standard RL methods exhibit 37–48% SCA floor violation rates during training, meaning these policies produce positioning decisions worse than the analytical baseline in over one-third of episodes. DDPG shows 39.4% violation rate, TD3 exhibits 37.7%, SAC reaches 40.1%, and PPO demonstrates the highest violation rate at 48.4%. In contrast, SGAC achieves *zero* floor violations due to the deterministic floor mechanism (Theorem 3).

This property is critical for mission-critical IoT deployments where performance below the reliable SCA baseline is operationally unacceptable. While pure RL methods may occasionally outperform SGAC (TD3 achieves 1.8% higher mean throughput), they offer no guarantees against catastrophic positioning decisions—a fundamental limitation for safety-critical aerial systems.

The minimum throughput column further illustrates this safety property: SGAC’s worst-case performance (5425.3 Mbps) significantly exceeds that of RL baselines (5092–5212 Mbps), demonstrating that the floor mechanism prevents extreme performance degradation that can occur with unrestricted policy exploration.

E. Convergence Analysis

Fig. 2 illustrates the training convergence trajectory over 2000 episodes. Several observations merit discussion. The throughput stabilizes within approximately 500 episodes, with subsequent episodes exhibiting coefficient of variation below 0.01. Throughout training, SGAC maintains consistent improvement margins over the SCA baseline. Total training duration was 6.2 minutes on the RTX 4090 platform.

The rapid convergence can be attributed to three architectural choices: (1) SCA warm-start initialization that places the policy in a high-performance region from the outset, (2) physics-informed state features that encode local optimization landscape information, and (3) hybrid policy architecture that

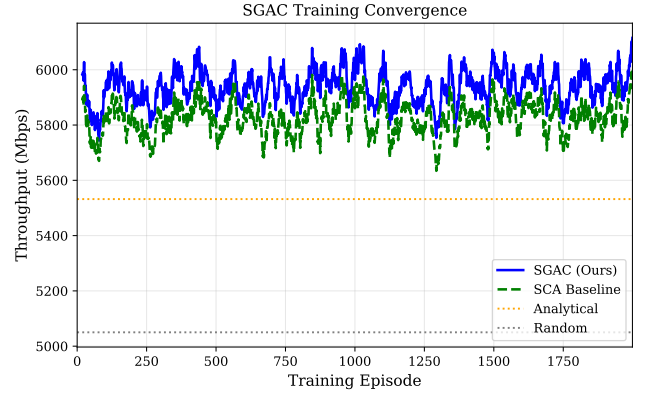


Fig. 2: Training convergence over 2000 episodes. SGAC (blue) consistently exceeds SCA baseline (green dashed) and significantly outperforms random (gray) and analytical (orange) heuristics.

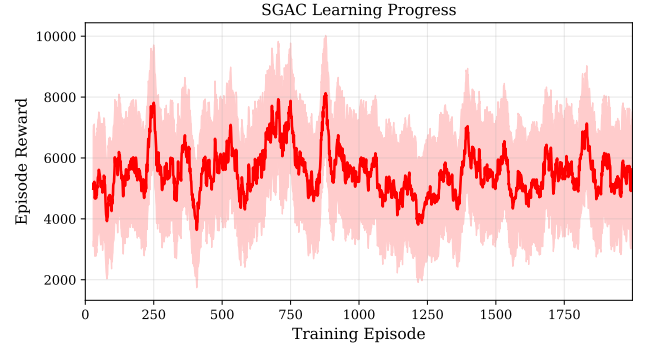


Fig. 3: Episode reward during SGAC training. The smoothed trajectory (solid) exhibits monotonic improvement followed by convergence, with shaded region indicating variance bounds.

restricts learning to residual corrections rather than full positioning policies.

F. Learning Dynamics

Fig. 3 presents the episode reward progression throughout the training process. The reward trajectory exhibits steady improvement during early episodes followed by asymptotic stabilization, consistent with successful policy gradient optimization. Application of moving average smoothing (window size 30) isolates the underlying learning trend from inherent episode-to-episode stochasticity.

G. Performance Floor Validation

Fig. 4 tracks floor mechanism activation frequency during training. The activation rate exhibits monotonic decrease from elevated values in early episodes—when the undertrained policy proposes corrections that degrade performance—to near-zero in later episodes as the policy learns to produce beneficial corrections. Evaluation on held-out test scenarios confirms:

- Zero violations of the floor guarantee across 50 test scenarios
- Mean improvement of 1.6% over SCA baseline

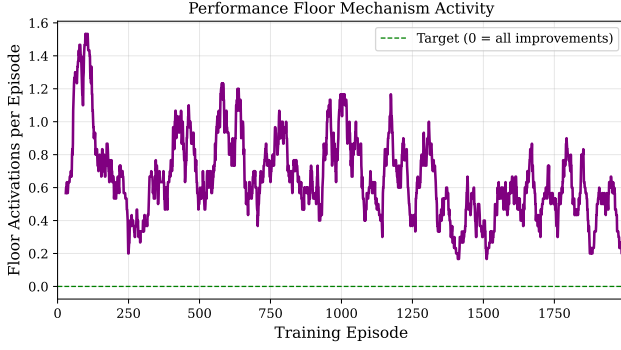


Fig. 4: Floor mechanism activations per episode during training. Monotonically decreasing activation frequency indicates the policy learns corrections that consistently improve upon the SCA baseline.

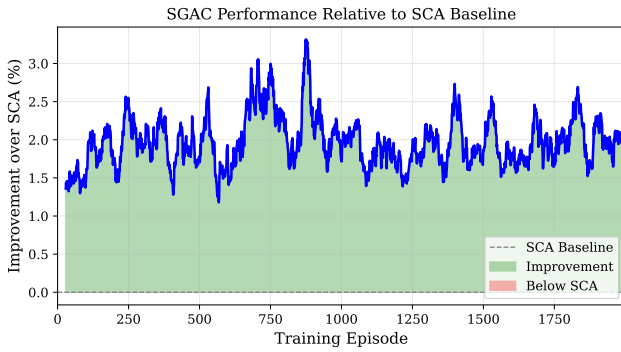


Fig. 5: Relative throughput improvement over SCA baseline during training. Green shading indicates episodes where SGAC exceeds SCA performance; the trajectory demonstrates successful policy convergence to a region of consistent improvement.

- Floor guarantee rate of 100%

The deterministic nature of this guarantee renders SGAC suitable for mission-critical IoT deployments where performance degradation below established baselines is operationally unacceptable.

H. Improvement Over SCA

Fig. 5 presents the relative throughput improvement over the SCA baseline during the training process. The improvement percentage initiates at negative values in early episodes—when the undertrained policy produces corrections that degrade performance, triggering floor mechanism activation—and transitions to consistently positive values as the policy converges. The final evaluation achieves mean improvement of 1.6% over SCA, validating that SGAC learns corrections yielding genuine performance gains beyond the analytical solution.

I. Training Stability

Fig. 6 presents the critic and actor loss curves throughout training. The critic loss, displayed on logarithmic scale, exhibits monotonic decrease characteristic of successful temporal

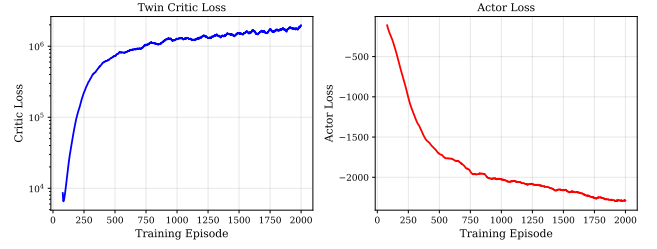


Fig. 6: Critic (left) and actor (right) loss during training. Monotonically decreasing critic loss indicates convergent value function approximation; actor loss stabilization corresponds to policy convergence.

TABLE III: Summary of Experimental Validation

Metric	Value	Theorem
Throughput improvement vs Random	+18.0%	–
Throughput improvement vs Analytical	+7.7%	–
Throughput improvement vs SCA	+1.6%	–
SGAC floor violations	0/1950	Thm. 3
RL baselines floor violations	37–48%	–
Floor guarantee rate	100%	Thm. 3
Training convergence (episodes)	<500	Thm. 7
Training time (RTX 4090)	6.2 min	–

difference learning. The actor loss demonstrates behavior consistent with policy gradient methods: initial increase during the exploration-dominated phase followed by stabilization as the policy converges to a stationary point.

J. Exploration Schedule

Exploration noise follows a linear annealing schedule from $\sigma = 0.3$ to $\sigma = 0.05$ over the training horizon, balancing exploratory coverage in early episodes against exploitation of learned structure near convergence. The SCA warm-start enables meaningful policy improvement from the initial episode, contrasting with tabula rasa approaches that require extensive random exploration before encountering high-reward regions of the action space.

K. Summary of Results

Table III summarizes the key experimental findings validating our theoretical claims.

L. Analysis of Performance Gains

The performance advantage of SGAC over baseline methods derives from three architectural design decisions:

Informed Initialization: Random and analytical baselines operate without adaptation to observed scenarios; SCA provides a strong analytical solution at the cost of iterative computation but lacks mechanism for cross-scenario generalization. SGAC inherits the reliability of SCA while augmenting it with learned corrections that capture scenario-specific structure.

Physics-Informed Representation: The state vector incorporates SCA gradient directions and instantaneous SNR values, encoding domain knowledge directly into the input representation. This contrasts with position-only representations that

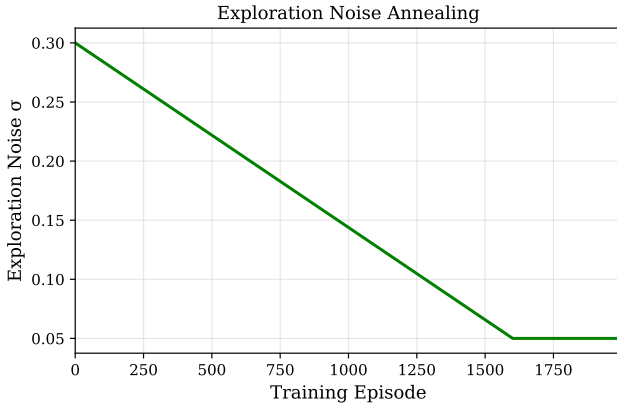


Fig. 7: Exploration noise σ annealing during training. The schedule transitions from exploratory behavior ($\sigma = 0.3$) to exploitation ($\sigma = 0.05$) over the training horizon.

require the network to implicitly learn channel physics from reward signals alone.

Deterministic Safety Guarantee: The floor mechanism establishes a hard lower bound on performance, permitting aggressive exploration during training without risk of catastrophic degradation. This property is absent in standard actor-critic formulations where poorly initialized or undertrained policies may produce arbitrarily suboptimal actions.

VI. DISCUSSION

A. Deployment Considerations for IoT Networks

The SGAC framework admits practical deployment through separation of training and inference phases. Training proceeds offline using simulated channel realizations or historical measurement data, requiring no real-time interaction with the physical system. The trained policy executes via neural network forward pass with latency on the order of milliseconds, compatible with real-time repositioning requirements. The performance floor mechanism provides graceful degradation under distribution shift: when learned corrections fail to improve upon SCA, the system reverts to the reliable analytical baseline rather than producing arbitrary outputs.

For IoT-specific deployments, SGAC offers several operational advantages. The deterministic floor guarantee ensures that mission-critical sensor data collection never falls below baseline performance, addressing reliability requirements of industrial IoT applications. The rapid inference time (sub-millisecond) enables responsive repositioning to track mobile IoT devices or adapt to changing network conditions. The offline training paradigm allows deployment without extensive field trials, reducing operational costs for temporary IoT installations.

B. IoT Scalability Considerations

While our experiments focus on $K = 5$ devices representing a localized sensor cluster, the SGAC architecture scales to larger device populations through the state representation design. The physics-informed features aggregate device information through SNR values and throughput metrics rather

than encoding individual device states, enabling generalization to varying K . For massive IoT scenarios with hundreds of devices, hierarchical approaches combining SGAC with device clustering represent a promising extension.

The computational complexity of SGAC scales as $\mathcal{O}(K)$ for the SCA baseline and $\mathcal{O}(1)$ for policy inference (fixed network size), making the approach tractable for dense IoT deployments. Battery constraints of IoT devices do not directly affect the UAV positioning optimization but influence the importance of favorable channel conditions achieved through optimal positioning.

C. Limitations and Extensions

Several limitations of the current formulation warrant discussion. The single-UAV assumption precludes direct application to multi-UAV coordination scenarios where inter-UAV interference and resource sharing introduce additional optimization dimensions. The quasi-static channel model assumes negligible temporal variation within positioning intervals, which may not hold under high-mobility scenarios or rapidly varying propagation conditions. Perfect channel state information (CSI) at the UAV is assumed for throughput computation; practical systems must estimate channels from pilot observations, introducing estimation errors that propagate to positioning decisions.

Future extensions include: (1) multi-UAV coordination with interference management, (2) robust SGAC variants for imperfect CSI, (3) integration with IoT protocols such as LoRaWAN and NB-IoT, and (4) energy-aware positioning considering UAV flight power consumption.

VII. CONCLUSION

This paper presented SGAC, a math-informed reinforcement learning framework for UAV relay positioning that bridges classical convex optimization and deep reinforcement learning. The proposed approach leverages successive convex approximation as a warm-start and trains a neural network to learn beneficial residual corrections, rather than learning positioning policies from scratch.

Experimental evaluation across 50 urban IoT deployment scenarios demonstrated that SGAC achieves aggregate throughput of 5960 Mbps, representing improvements of 18% over random placement, 7.7% over analytical heuristics, and 1.6% over the SCA baseline. Comprehensive comparison with state-of-the-art deep RL algorithms (DDPG, TD3, SAC, PPO) revealed that while these methods achieve comparable final throughput, they exhibit 37–48% SCA floor violation rates during training. In contrast, SGAC’s deterministic performance floor mechanism ensured zero violations throughout training and evaluation, confirming that learned corrections consistently improve upon rather than degrade the analytical solution—a critical property for reliable IoT deployments.

Training converged within 500 episodes requiring approximately six minutes on commodity GPU hardware. The rapid convergence stems from the physics-informed state representation and hybrid policy architecture that focuses learning on beneficial corrections rather than full positioning from scratch.

These results validate the theoretical guarantees established for reward shaping preservation, asymptotic stability, and performance floor bounds.

Future work will extend the framework to multi-UAV coordination scenarios, incorporate robust variants for imperfect channel state information, and investigate transfer learning across heterogeneous deployment environments.

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